

PLANT DISEASE CLASSIFICATION

PlantVillage Dataset • Technical Dataset Brief

54,304 images • 38 classes • 14 crop species • 21 disease labels

Dataset: [geraldmc/plantvillage-full](#)

Revision v0.1.0

16:9 presentation format



1. Dataset Overview

The project uses the curated Hugging Face release geraldmc/plantvillage-full, revision v0.1.0. It packages the color PlantVillage images with structured metadata and a leaf-grouped train/test assignment.



Dataset properties

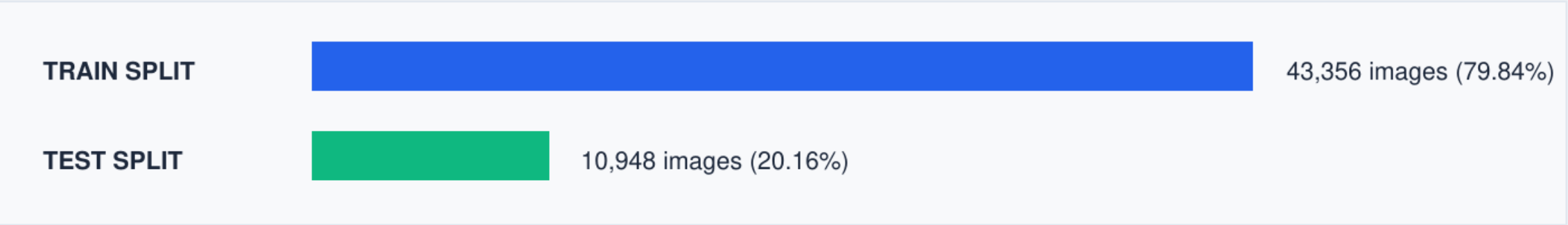
Dataset	PlantVillage (full)
Hugging Face release	geraldmc/plantvillage-full
Revision	v0.1.0
Rows / images	54,304
Channels	RGB (3 channels)
Approx. size	~855 MB

Original PlantVillage context

The original collection is described by Mohanty, Hughes & Salathé (2016) as 54,306 images from 14 crop species and 26 disease conditions, represented as 38 crop–disease classes. This project reports the 54,304 images actually loaded.

2. Dataset Structure & Split

Hugging Face exposes one underlying split named train. The actual train/test assignment is fully preserved inside the split metadata column.



The split incorporates a strict **leaf-grouped** configuration, guaranteeing that images belonging to the same physical leaf do not leak across validation boundaries.

3. Class Distribution (Sample Overview)

Substantial imbalance exists across the 38 crop–disease combinations:

1. Apple — healthy (1,645)	7. Grape — Black rot (1,180)	13. Potato — Early blight (1,000)
2. Blueberry — healthy (1,502)	8. Grape — Esca (1,383)	14. Potato — Late blight (1,000)
3. Cherry — Powdery mildew (1,052)	9. Grape — Leaf blight (1,076)	15. Soybean — healthy (5,090)
4. Corn — Common rust (1,192)	10. Orange — HLB / Greening (5,507)	16. Squash — Powdery mildew (1,835)
5. Corn — Northern Leaf Blight (985)	11. Peach — Bacterial spot (2,297)	17. Tomato — Yellow Leaf Curl (5,357)
6. Corn — healthy (1,162)	12. Pepper, bell — healthy (1,477)	18. Tomato — healthy (1,591)

* Complete overview covers all 38 classes with a total dataset size of 54,304 items.

4. Crop & Disease Coverage

The dataset spans across a robust cross-section of global agriculture, mapping 14 host crop species against 21 disease conditions or healthy labels.

14 Host Species: Apple • Blueberry • Cherry (including sour) • Corn (maize) • Grape • Orange • Peach • Pepper, bell • Potato • Raspberry • Soybean • Squash • Strawberry • Tomato

21 Disease / Condition Labels: Apple scab • Black rot • Cedar apple rust • Healthy • Powdery mildew • Cercospora leaf spot • Gray leaf spot • Common rust • Northern Leaf Blight • Esca (Black Measles) • Leaf blight • HLB (Citrus greening) • Bacterial spot • Early blight • Late blight • Leaf scorch • Leaf Mold • Septoria leaf spot • Spider mites • Target Spot • Yellow Leaf Curl Virus • Tomato mosaic virus

5. Sample Leaf Images



Potato leaf examples | Healthy • Early blight • Late blight

Representative image supplied from the PlantVillage dataset

6. Reproducible Dataset Loading

```
from datasets import load_dataset
ds = load_dataset("geraldmc/plantvillage-full",
                  revision="v0.1.0")
```

No raw dataset files are stored in the GitHub project. The notebook downloads the dataset from Hugging Face at runtime.

Source	Hugging Face	Release	v0.1.0	Image	PIL, 256×256 RGB
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7. Limitations & Interpretation

- 1 **Controlled Environment:** Images use plain backgrounds and controlled acquisition conditions, which are simpler than typical field photographs.
- 2 **Domain Shift:** Strong PlantVillage test performance should not automatically be interpreted as equivalent real-world field performance.
- 3 **Leaf Grouping:** The leaf-grouped split helps prevent images of the same physical leaf from appearing across train and test sets.
- 4 **Class Imbalance:** Overall accuracy should always be accompanied by precision, recall, F1-scores, and per-class metrics.

Interpretation Rule: Treat the dataset as a strong controlled benchmark for image classification—not as a substitute for diverse real-world field imagery.

8. References

Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016).

Using Deep Learning for Image-Based Plant Disease Detection. *Frontiers in Plant Science*, 7, 1419.

PlantVillage Dataset — original GitHub repository

[spMohanty/PlantVillage-Dataset](#) — dataset overview and sample image paths.

Project dataset release

[geraldmc/plantvillage-full](#), Hugging Face Datasets, revision v0.1.0.

TensorFlow Datasets catalog

PlantVillage — 54,303 images, 38 categories by species and disease.

Web-verified sources

[github.com/spMohanty/PlantVillage-Dataset](#) • [huggingface.co/datasets/geraldmc/plantvillage-full](#) • [tensorflow.org/datasets/catalog/plant_village](#)